

PROJECT REPORT ON

Click & Conquer : Unveiling the B2C E-Commerce Odyssey of Vadodara City

A REPORT SUBMITTED TO



DEPARTMENT OF STATISTICS
FACULTY OF SCIENCE
THE MAHARAJA SAYAJIRAO UNIVERSITY OF BARODA

YEAR – 2025

GUIDE:

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**PROJECT
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DECLARATION

We hereby declare that the work presented in this project titled "Click & Conquer : Unveiling the B2C E-Commerce Odysseys of Vadodara City" is the result of our independent work, submitted to the Department of Statistics, at The Maharaja Sayajirao University Of Baroda, as a part of our journey towards earning our Bachelor of Science degree, we undertook an exciting project under the expert guidance of Prof. (Dr.) Manisha Pandya and Priti Singh Mam.

This project showcases our original research efforts and represents a significant milestone in our academic journey. Importantly, the findings presented here have not been shared with any other University or Institute.

Place: Vadodara

CERTIFICATE

This is to certify that Priyanshu Garg, Jaymeet Modi, Bhagyashree, Jatin, Vasundhara have successfully and satisfactorily completed the project titled:

“Click & Conquer : Unveiling the B2C E-Commerce Odyssey of Vadodara City”

As a team for the academic year 2025-26 have submitted the work to the Department of Statistics as a partial fulfilment of the requirement of the Bachelor's Degree in Statistics and have represented their original work under the supervision and guidance of Prof. (Dr.) Manisha Pandya and Preeti Singh.

I wish them grand success in the future

Prof. V .A. Kalamkar
Head of Department

Prof.(Dr.) Manisha Pandya
Priti Singh

ACKNOWLEDGMENT

First and foremost, we would like to thank our guide, Dr. Mansiha Pandya and Priti Singh Mam who always helped us, welcomed our questions, kept us motivated and gave us a lot of recommendations and suggestions. We would not have reached this phase, if it were not for his permanent support, advice, and guidance.

We also would like to express our gratitude to our Head Dr. Vipul Kalamkar for their support, guidance and helpful feedback.

We would also thank our university to provide us with such great faculties, library, environment to grow ourselves, and for the platform to explore ourselves and showcase our skills.

Our sincere thanks to our parents for guiding us decently and supporting us at every stage in our life also their wishes for successful completion of this project.

Finally, we would like to express our eternal gratitude to our seniors and friends for their support, appreciation and patience. We would like to dedicate this report to them all.

ABSTRACT

This study explores diversity in online shopping within Vadodara City by understanding preferences across different neighbourhoods. This presentation looks at how online shopping have evolved and the plausibility of upcoming future technologies.

This study presents a set of experiments that involve the use of R Programming for computation, M.S Excel for Data Processing and Visualization, Sampling procedure, Chi-Square test of Independence as well as various Categorical Analysis Techniques and Non-Parametric Testing.

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INTRODUCTION

Think of it like a journey—from scrolling through websites to finally receiving that exciting delivery at your doorstep. How do people in Vadodara, across different age groups and areas, interact with e-commerce platforms? What influences their choices, and how has their shopping behavior evolved in the digital age?

As we take a closer look, we'll see that online shopping is no longer limited to just the tech-savvy youth. From students buying gadgets to homemakers ordering kitchen essentials, e-commerce has made its way into almost every household. Leading platforms like Amazon, Flipkart, and Meesho are shaping buying patterns through features like fast delivery, product reviews, and attractive offers.

However, the journey isn't always smooth. Some users still face issues—slow websites, confusing checkout steps, or poor return policies. These small but significant barriers often impact user satisfaction and their decision to complete a purchase.

Ultimately, this project seeks to unveil the B2C e-commerce odyssey in Vadodara by analyzing how people shop, what drives their preferences, and how their experiences can be improved for a smarter, smoother digital shopping future.

LITERATURE REVIEW

Understanding the evolving landscape of B2C e-commerce in India, especially in tier-2 cities like Vadodara, is vital in today's consumer-driven digital economy. Various research studies have highlighted key behavioral trends, technological enablers, and challenges shaping online shopping patterns among Indian consumers.

Kotler & Keller (2016) emphasized the increasing tendency of consumers to compare products and rely on digital feedback before making purchases. Their research concluded that online reviews, platform reputation, and accessibility strongly affect purchase intentions.

Agarwal & Rastogi (2021) explored consumer trust in e-commerce platforms and found that transparent return policies, timely delivery, and responsive customer service are primary factors influencing repeated purchases and brand loyalty in the Indian context.

Chaffey (2019) analyzed the significance of web design and user experience. The study showed that intuitive navigation, faster website loading speed, and optimized mobile experiences greatly enhance user satisfaction, reducing cart abandonment rates.

Gupta & Sharma (2020) studied the impact of digital payment integration in e-commerce, concluding that platforms offering multiple secure payment options witnessed higher conversion rates. Security concerns and lack of trust in new payment methods remained notable hurdles.

OBJECTIVES OF STUDY

- 1)To understand the demographic profiles of online shoppers within the target population.
- 2)To identify the leading e-commerce platforms, key product preferences, and drivers behind consumer choices.
- 3)To examine the relationship between demographic factors and online shopping preferences using the Chi-square test.
- 4)To assess the association between demographic factors and online shopping preferences using Goodman and Kruskal's test and CDA.
- 5)To conduct a SWOT analysis of the e-commerce landscape, specifically focusing on Baroda's population.

METHODOLOGY AND DATA

In this project, we were primarily focused on exploring online shopping trends, technology, and user preferences in Vadodara. The type of data and the variables we were interested in studying were not readily available in the form of the secondary dataset, and hence, we as the team decided to collect **primary data** and analyze it with our specific goals in mind.

We have designed **17 questions** around the Likert scale, which helped us ask clear, categorical questions.

This scale made it easy for people to express their views on payment trends, technology, and user preferences in Vadodara. In total, we received **323 responses**.

With our questionnaire being quite comprehensive, we began by cleaning up the dataset.

One of the key objective was obtaining a comprehensible data, especially for our latest project, which relies on primary sources. Making sure our data is accurate has been a big job. Along the way, we've learned a ton and faced lots of challenges. From figuring out how to collect primary data effectively to discovering new things, it's been a great learning experience for us.

When it comes to collecting primary data, the spotlight shines brightest on questionnaire preparation. Each question needed to be clear, to the point, and easy to understand. This helps in getting the most accurate and honest answers from respondents. Also, it becomes easier to clean the data, as there are very few absurd entries.

TYPE OF DATA:

PRIMARY DATA: It is a first-hand original piece of work collected by the researcher by the objective of their study. It has been collected through well-framed self-administered questionnaires through surveys and schedules where necessary by circulating hard and soft copies. Data is collected from the students, salaried persons, business persons, homemakers, and other left categories of occupation of the selected city.

SAMPLING METHODS:

Sampling methods are structured techniques used to pick a smaller group from a larger population for research. The methodology used to sample from a larger population depends on the type of analysis being performed.

SAMPLING PROCEDURE:

We employed Convenience Sampling but in order to increase the representativeness of the sample we incorporated it with Area sampling by dividing the map of Vadodara City into four quadrants with the help of grid lines, creating 25 blocks in each quadrant, and selecting five blocks from each quadrant for data collection.

SAMPLE SIZE DETERMINATION:

Sample size is the number of observations or measurements taken in a study, influencing the reliability and precision of research findings. It's a critical consideration in research design, with larger sample sizes generally leading to more accurate estimates and stronger statistical power for detecting effects or relationships of interest.

- Target Age Group : 18 and above

Formula for calculating large sample size –

$$\text{Sample size}(n) = \frac{z^2 p (1-p)}{ME^2}$$

Where,

Z= Z-score = 1.96

p = Sample proportion = 22/64 = 0.3

M.E= Margin of error =0.05

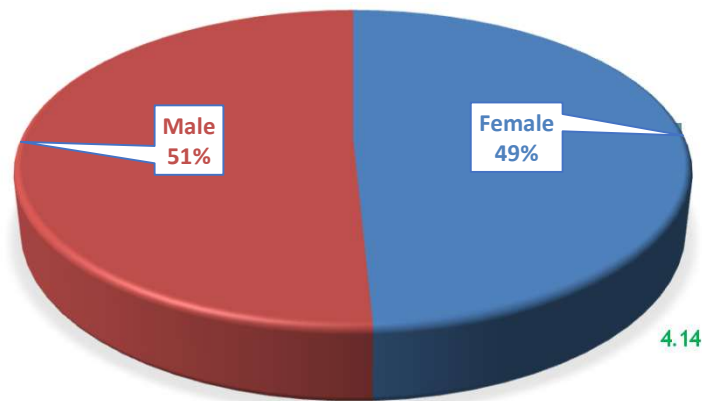
α = 5%

n = 323

So our determined sample size is 323.

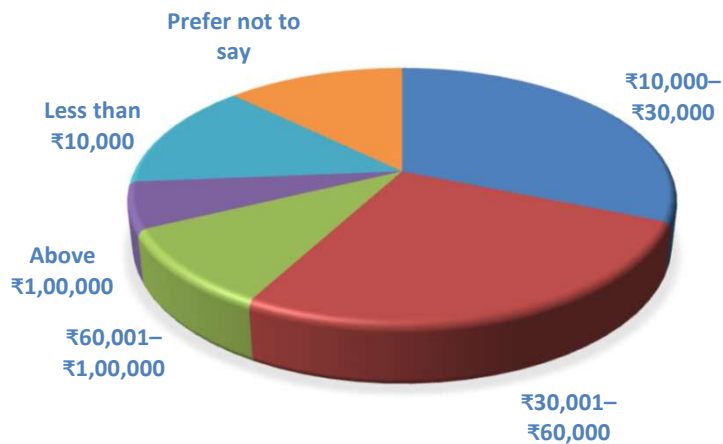
EXPLORATORY DATA ANALYSIS

GENDERWISE COUNT

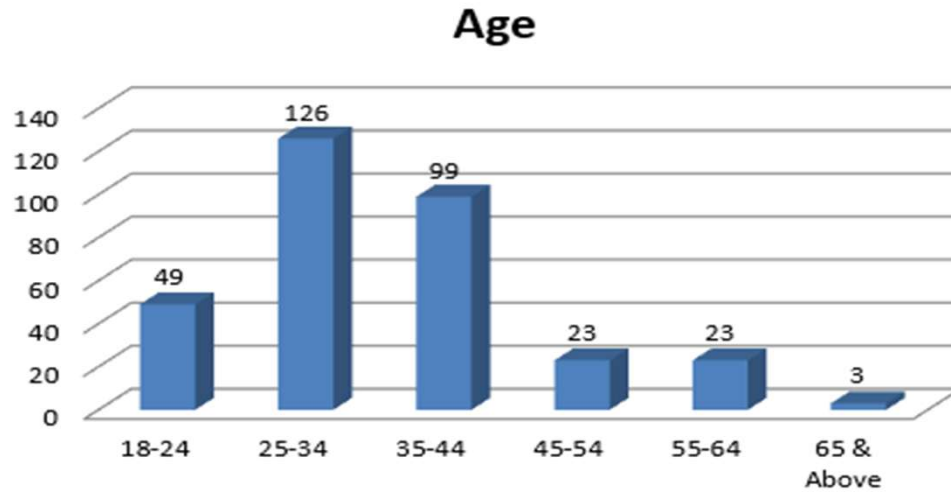


Interpretation: The gender distribution is almost evenly split between male and female respondents, with a slight majority of males (about 1.6% more).

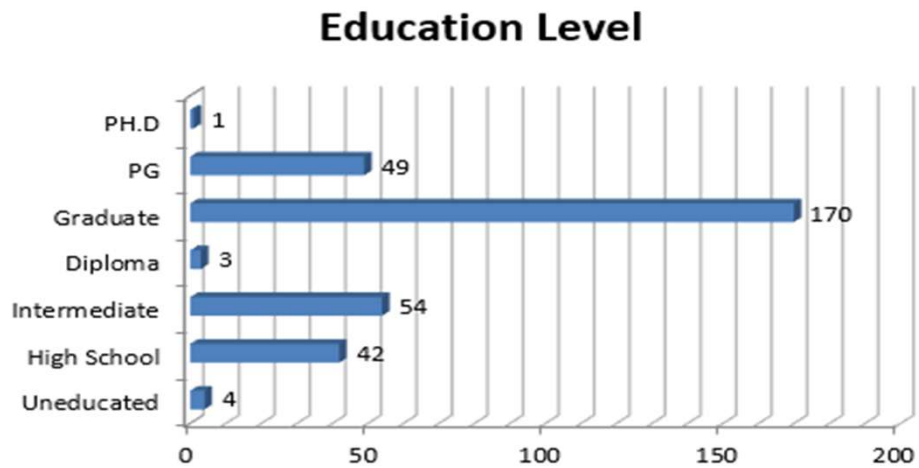
INCOME LEVEL



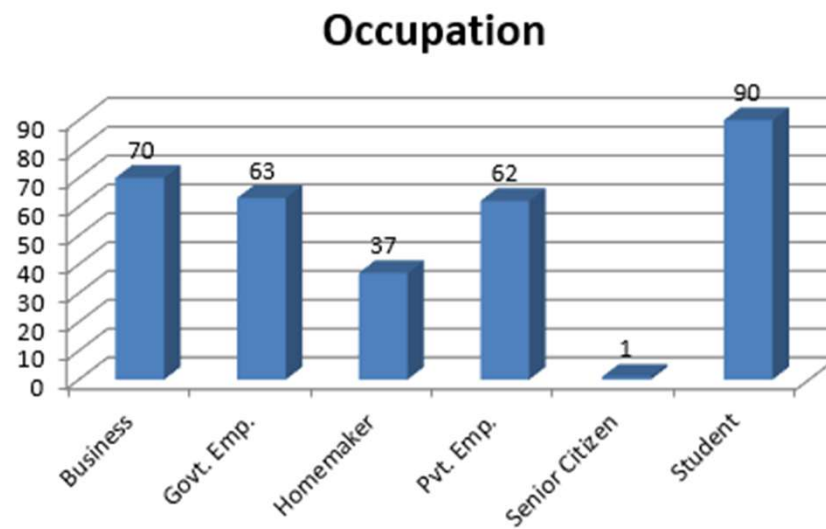
Interpretation: The majority of the respondent are fall in the 10000 to 30000 30000 to 60000 income brackets.



Interpretation: The majority 126 out of 323 respondents belong to the age group of 18 to 24.

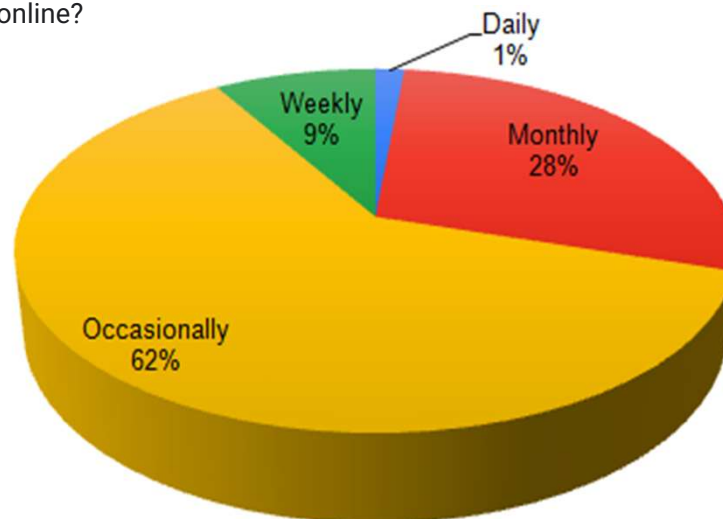


Interpretation: The majority 170 out of 323 respondents are Graduates.



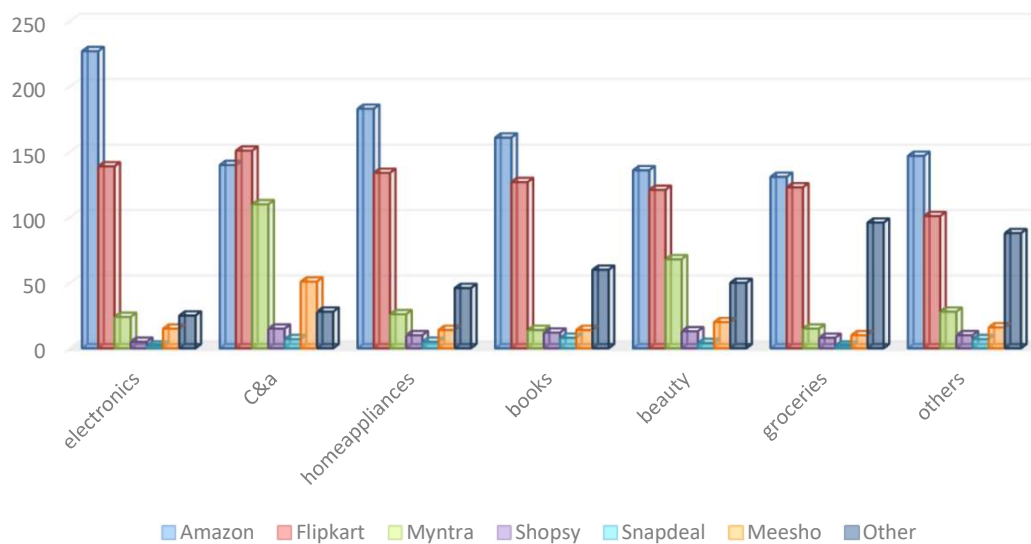
Interpretation: The majority 90 out of 323 respondents are student and then followed by a Businessman etc.

How often do you shop online?



Interpretation: Majority shop occasionally – this reflects a medium level of engagement with online shopping platforms.

What type of products and in which e-commerce platform do you use frequently for purchasing?



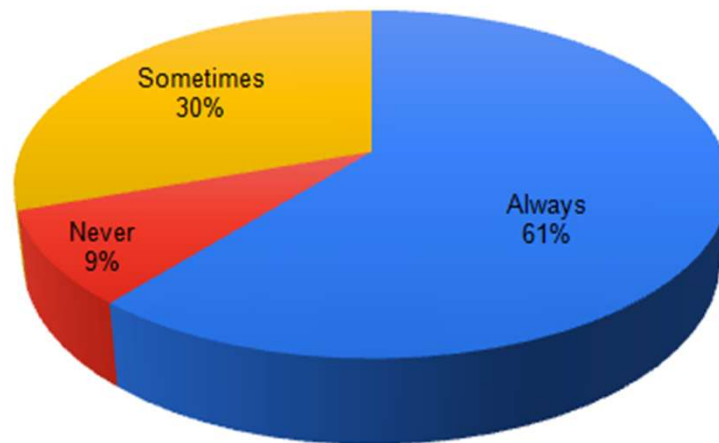
Interpretation: Amazon is the top choice across most categories, especially for Electronics, Home Appliances, and Books.

When choosing an e-Commerce platform, how important are the following factors for you?



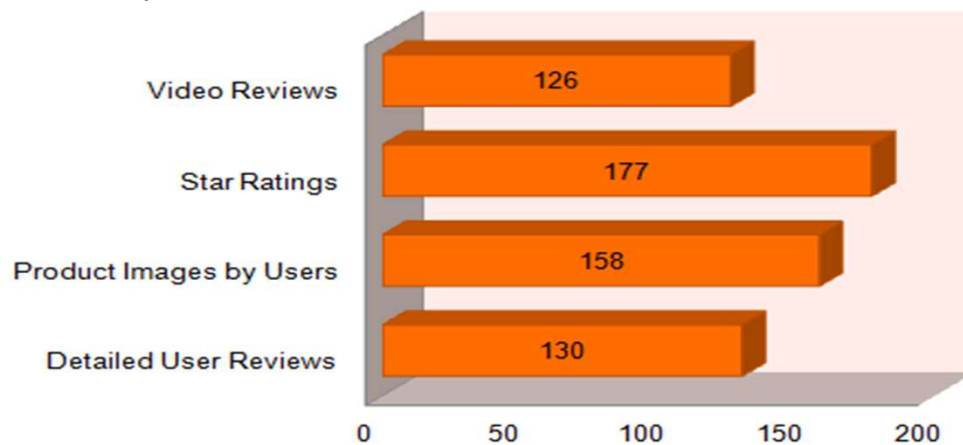
Interpretation: Shoppers prioritize quality, when picking a platform. If we talk about product quality and variety 105 respondents choose this as a most important.

How often do you read reviews and ratings before online shopping?



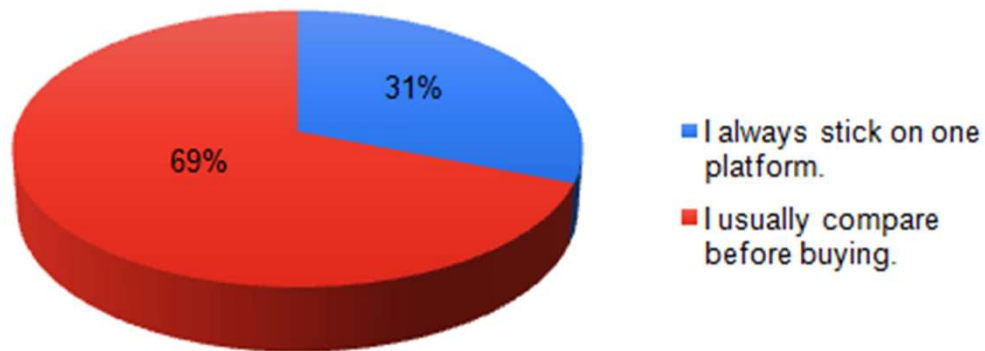
Interpretation: The vast majority read reviews always or most of the time.

What type of feedback do you find most useful? `



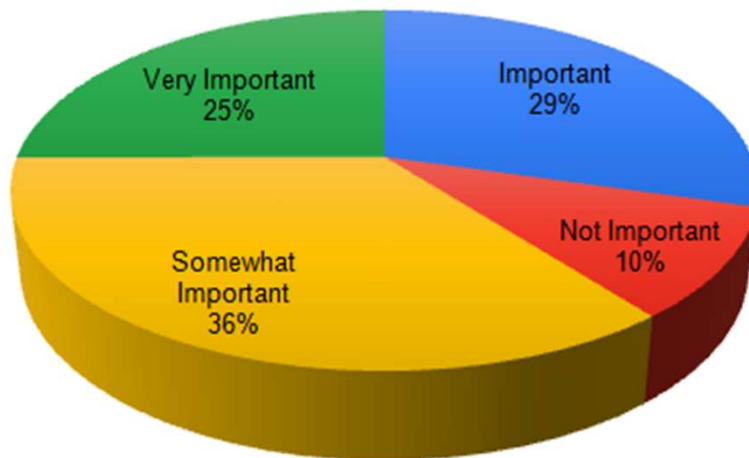
Interpretation: Shoppers prefer quick, visual feedback like star ratings and user-submitted images, but still appreciate detailed reviews for making informed decisions.

Do you compare before deciding, where to buy?



Interpretation: An overwhelming majority compare options before buying, underscoring the importance of competitive pricing and product differentiation.

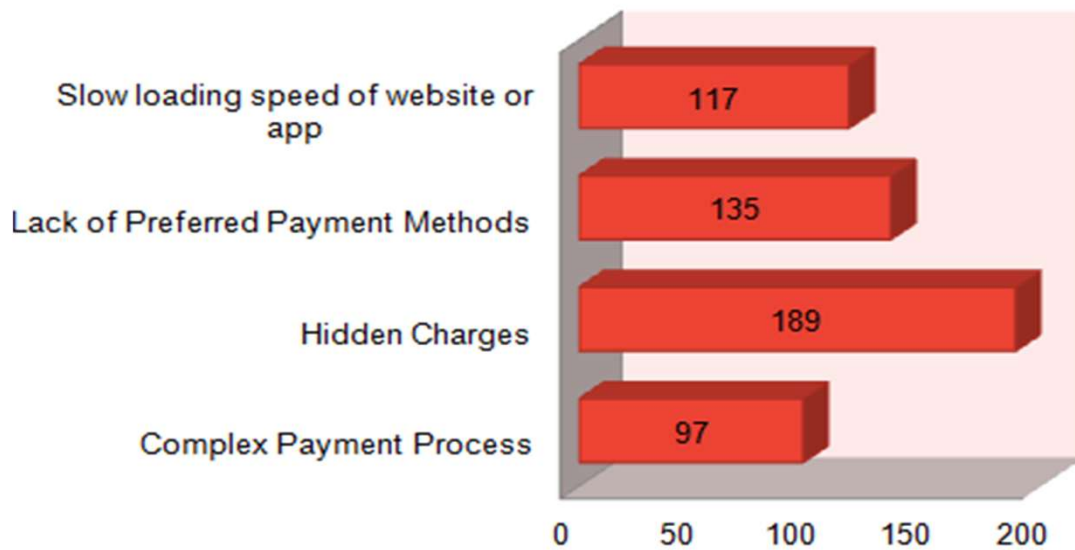
How important is website design in your online shopping experience?



Interpretation: Most respondents rated website design as important to very important.

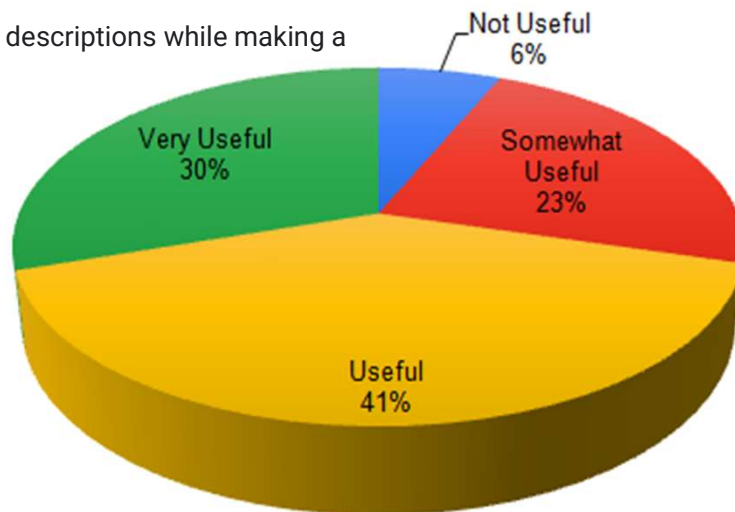
Aesthetics, usability, and user experience heavily influence customer trust and navigation ease.

What checkout barriers discourage you the most?



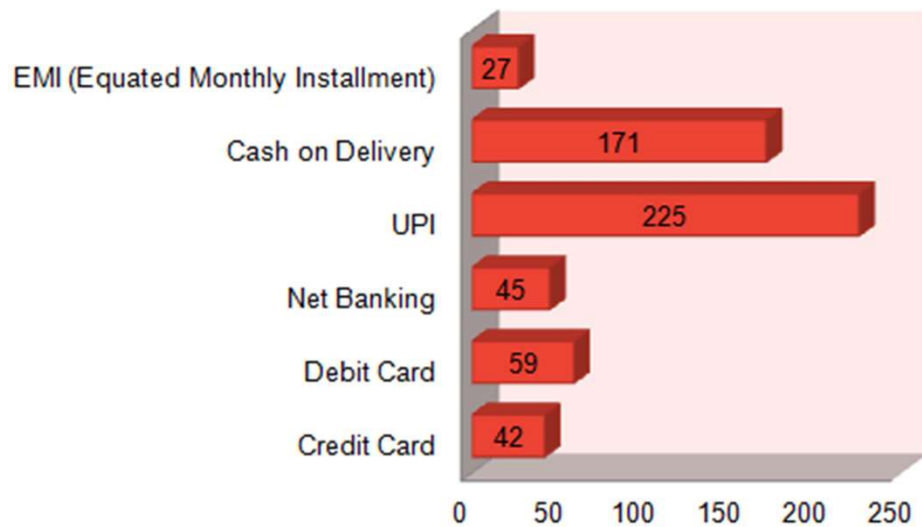
Interpretation: Shoppers dislike surprise costs the most, followed by inconvenient or slow payment experiences. Simplifying checkout and showing full costs upfront are key to retaining users.

How useful are product descriptions while making a purchase online?

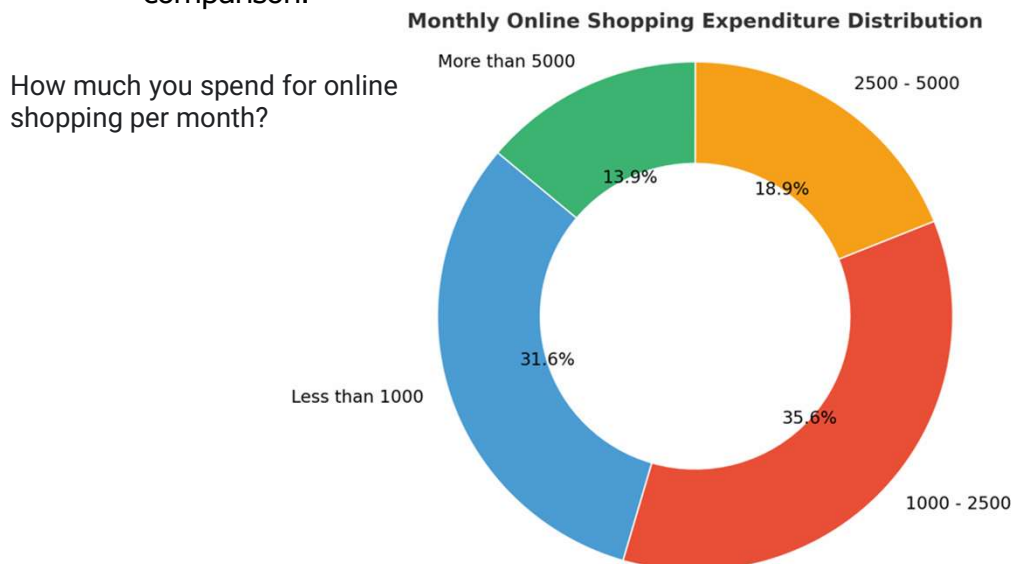


Interpretation: Highly useful to most respondents, indicating they rely on product descriptions to understand the item thoroughly before buying.

Which payment method do you use most often?



Interpretation: UPI leads in convenience and adoption, while COD still holds strong as a fallback for many. Cards and EMI are used less frequently in comparison.



Interpretation: Spending ranges vary, but most fall into mid-range spending categories. This provides an opportunity for e-commerce platforms to tailor promotions or product ranges based on different spending brackets.

STATISTICAL ANALYSIS

HYPOTHESIS TESTING :

Hypothesis testing is a statistical method used to determine if there's enough evidence in a sample to infer something about a population. It involves making educated guesses, comparing observed data to what's expected by chance, and deciding whether the results are statistically significant.

CHI-SQUARE TEST OF INDEPENDENCE

A chi-square test is a statistical test used to compare observed results with expected results. The purpose of this test is to determine if a difference between observed data and expected data is due to chance, or if it is due to a relationship between the variables you are studying.

GOODMAN KRUSKAL GAMMA TEST

The Goodman-Kruskal gamma test assesses the strength and direction of association between two ordinal variables. It measures the degree of concordance between rankings, disregarding ties, providing a non-parametric measure of correlation. Often used when variables lack a linear relationship, it quantifies ordinal associations beyond chance, aiding in inferential analysis.

CHI-SQUARE TEST OF INDEPENDENCE

Hypothesis-1

Ho: Age Group and Shopping Frequency are independent.

H1: Age Group and Shopping Frequency are not independent.

Contingency Table :

	Frequently	Occasionally	Row Total
18-24	22	27	49
25-34	49	77	126
35-44	38	61	99
45-54	8	16	24
55-64	7	15	22
65 & above	0	3	3
Col Total	124	199	323

Variables

X: Age Group

Y: Shopping

Frequency

Output

X-squared= 3.0901

Degrees of freedom=5

T-value= 11.0705

Interpretation: Since X-squared < T-value, we will accept our Null Hypothesis at 5% level of significance.

Hence, Age and Shopping Frequency are independent.

Hypothesis-2

Ho : Educational Qualification and Shopping Frequency are Independent.

H1: Educational Qualification and Shopping Frequency are not Independent.

Contingency Table :

	Frequently	Occasionally	Row Total
Uneducated	1	3	4
High School	18	24	42
Intermediate	20	34	54
Diploma/VC	0	3	3
Graduate	59	111	170
Postgraduate	26	23	49
Ph.D.	0	1	1
Col Total	124	199	323

Variables

X: Educational Qualification

Y: Shopping Frequency

Output

X-squared=8.6313

Degrees of freedom=6

T-value= 12.5916

Interpretation: Since X-squared < T-value, we will accept our Null Hypothesis at 5% level of significance.

Thus, Educational Qualification and Shopping Frequency are independent.

Hypothesis-3

Ho: Income Group and Shopping Frequency are Independent.

H1: Income Group and Shopping Frequency are Not Independent.

Contingency Table :

	Frequently	Occasionally	Row Total
N/A	16	25	41
Very Low	14	30	44
Low	41	60	101
Medium	30	56	86
High	13	19	32
Very High	10	9	19
Column Total	124	199	323

Variables

X:Income Group

Y: Shopping

Frequency

Output

X-squared= 3.1626

Degrees of freedom=5

T-value= 11.0705

Interpretation: Since X-squared < T-value, we will accept our Null Hypothesis at 5% level of significance.

Thus, Income Group and Shopping Frequency are **independent**.

Goodman-Kruskal Test of Association

"Goodman-Kruskal Gamma Test for Ordinal Categorical Variables" is an alternative to the Chi-Square Test of association and is used to test whether there exists a significant association between two ordinal variables or not.

Gamma : The Goodman-Kruskal Gamma value lies between -1 and +1. The nature of the association is interpreted by the sign (negative or positive association) and the strength is given by the absolute value of gamma as follows:

0.00 – 0.10	Negligible
0.10 – 0.20	Weak
0.20 – 0.40	Moderate
0.40 – 0.60	Relatively Strong
0.60 – 0.80	Strong
0.80 – 1.00	Very Strong

R-Code for GK Test

```
{r}
install.packages("MESS")
library(MESS)
gkgamma(matrix1, conf.level = 0.95)
```

Hypothesis-1

Ho: Gender and Website Design Importance have no association.

H1: Gender and Website Design Importance have association.

Contingency Table:

Row Labels	Not Im.	Some. Im.	Important	Very Impo.	Row Total
Female	10	62	51	36	159
Male	22	53	44	45	164
Column Total	32	115	95	81	323

Variables

X: Gender

Y: Website Design

Importance

Output

Gamma Value= -0.0128

P-value= 0.8811

Interpretation: Since $p\text{-value} > 0.05$, we will accept our Null Hypothesis at 5% level of significance.

There is no significant association between gender and the importance placed on website design in the online shopping experience. The gamma value is very close to zero, indicating no meaningful relationship

Hypothesis-2

Ho: Gender and Product Description Importance have no association.

H1: Gender and Product Description Importance have association.

Contingency Table:

Row Labels	Not Us.	Some. Us.	Useful	Very Use.	Row Total
Female	11	40	54	54	159
Male	10	34	77	43	164
Column Total	21	74	131	97	323

Variables

X: Gender

Y: Product

Description

Importance

Output

Gamma Value= -0.0273

P-value= 0.7569

Interpretation: Since $p\text{-value} > 0.05$, we will accept our Null Hypothesis at 5% level of significance.

There is no significant association between gender and the perceived usefulness of product descriptions. The gamma value is very close to zero, indicating no meaningful relationship.

Hypothesis-3

Ho: Income Group and Monthly Spending have no association.

H1: Income Group and Monthly Spending have association.

Contingency Table :

	<1000	1000-2500	2500-5000	>5000	Row Total
N/A	20	9	8	4	41
Very Low	21	15	5	3	44
Low	30	39	21	11	101
Medium	18	40	15	13	86
High	8	8	6	10	32
Very High	5	4	6	4	19
Column Total	102	115	61	45	323

Variables

X: Income Group

Y: Monthly

Spending

Output

Gamma Value = 0.23993

P-value= 0.0001474

Interpretation: Since $p\text{-value} < 0.05$, we will reject our Null Hypothesis at 5% level of significance.

There is significant association between gender and the monthly spending habits and also indicating moderate meaningful relationship.

Hypothesis-4

Ho: Age and Reviews Reading have no association.

H1: Age Group and Reviews Reading have association.

Contingency Table :

	Never	Sometimes	Always	Row Total
18-24	8	22	19	49
25-34	7	39	80	126
35-44	5	23	71	99
45-54	4	7	13	24
55-64	3	6	13	22
65 & Above	1	1	1	3
Column Total	28	98	197	323

Variables

X: Age Group

Y: Monthly Spending

Output

Gamma Value = 0.14739

P-value= 0.07901

Interpretation: Since $p\text{-value} > 0.05$, we will accept our Null Hypothesis at 5% level of significance.

There is no significant association between age and the reviews reading habits and also indicating weak meaningful relationship.

CATEGORICAL DATA ANALYSIS

CATEGORICAL DATA:

Categorical variables represent types of data that may be divided into groups. Categorical measurements are expressed in terms of natural language descriptions, but not in terms of numbers.

Sometimes categorical data can take numerical values, but those numbers do not have mathematical meaning.

CONTINGENCY TABLE:

A contingency table is a tabular representation of the frequency distribution of two or more categorical variables. It shows the number of observations that fall into each combination of categories for the variables being studied. Contingency tables are used to explore relationships and dependencies between categorical variables in statistical analysis.

PEARSON'S STANDARDIZED RESIDUALS:

Pearson residuals are used in a Chi-Square Test of Independence to analyse the difference between observed cell counts and expected cell counts in a contingency table.

PARTIAL ODDS RATIO

The following is a 2x2x6 Contingency table. Independent Variable(X) :

Gender

Dependent Variable(Y) : Shopping Frequency

Stratification Variable(Z) : Age-Group

Shopping Frequency (Y)				
Age-Groups(Z)		Freque.	Occasi.	Odds Ratio
	Gender(X)			
18-24	Female	13	12	1.80
	Male	9	15	
25-34	Female	18	30	0.90
	Male	31	47	
35-44	Female	20	40	0.58
	Male	18	21	
45-54	Female	4	9	0.77
	Male	4	7	
55-64	Female	4	8	1.16
	Male	3	7	
65 & above	Female	0	1	1.66
	Male	0	2	

Interpretation: From the **Partial Contingency Tables** we can observe that for age groups 25-34 to 45-54, the odds of males doing online shopping frequently are higher than that of females. However for the age group 18-24, 55-64, 65 & Above the odds of females doing online shopping frequently are higher than that of males.

Pearson's Standardized Residuals

Pearson's Standardized Residual is given by the following formula :

n_{ij} =observed frequency for (i,j)th cell

u_{ij} =expected frequency for (i,j)th cell

1. Association between Occupation(X) and Shopping Frequency (Y)

$$\frac{n_{ij} - \hat{\mu}_{ij}}{[\hat{\mu}_{ij}(1 - p_{i+})(1 - p_{+j})]^{1/2}}$$

	Frequ.	Occasi.	Row Total	Pi.
Business	32	38	70	0.22
Govt. Empl.	25	37	62	0.19
Homemaker	13	24	37	0.11
Pvt. Empl.	25	38	63	0.19
Sr. Citizen	0	1	1	0.0031
Student	29	61	90	0.28
Col Total	124	199	323	1
P.j	0.38	0.62	1	

n_{ij}	u_{ij}	Resi.
32	26.87	1.42
38	43.13	-1.43
25	23.80	0.35
37	38.19	-0.35
13	14.20	-0.43
24	22.79	0.43
25	24.18	0.23
38	38.81	-0.23
0	0.38	-0.79
1	0.62	0.79
29	34.55	-1.41
61	55.45	1.42

Interpretation : Businessman, Government Employee, Private Employee are doing online shopping frequently.
While Homemaker, Senior Citizen, Student are doing online shopping occasionally.

2. Association between Localities with Gender (X) and Shopping Frequency (Y)

	Frequ.	Occas.	Row Total	pi.
Female	59	100	159	0.49
Male	65	99	164	0.51
Column Total	124	199	323	1
p.j	0.38	0.62	1	

nij	uij	Residuals
59	61.04	-0.46
100	97.96	0.46
65	62.96	0.46
99	101.04	-0.47

Interpretation: Male are doing online shopping frequently.
While Female are doing online shopping occasionally.

SWOT ANALYSIS OF ONLINE SHOPPING

STRENGTH:

- 1) **Diverse and Digitally Active Demographics** – In Our survey among Vadodara's urban population shows high adaptability to e-commerce, especially among youth and working professionals.
- 2) **Popular Platforms with Strong Local Penetration** – Amazon, Flipkart, and Meesho dominate local preferences, offering tailored deals and fast delivery to Baroda-specific PIN codes.
- 3) **Strong Influence of Reviews and Ratings** – Consumers in Vadodara actively consider reviews, reflecting digital literacy and research-driven buying behavior (supports Objective 3).
- 4) **Growing Trust in Online Shopping** – Return policies, CoD, and local warehouse availability increase confidence, leading to rising B2C engagement.

WEAKNESSES:

- 1) Digital Divide within Demographics** – Older adults and residents in peripheral zones of Vadodara still hesitate to adopt online shopping due to unfamiliarity or language barriers.
- 2) Overloaded or Poor Web Design** – Many respondents highlighted complex navigation and poor product filtering as a deterrent (directly from Objective 4).
- 3) Misleading Product Descriptions** – A mismatch between online product images and actual items received contributes to dissatisfaction (Objective 4 insight).
- 4) Dependence on Star Ratings Alone** – Some consumers rely solely on star ratings without reading detailed reviews, leading to impulsive or regrettable purchases.

OPPORTUNITIES:

- 1) Customized Local Campaigns Based on Consumer Insights** – Your study shows local product preferences and shopping motivations, which can help platforms target Baroda's population more effectively.
- 2) Improving Checkout UX (User Experience)** – Simplifying checkout processes and reducing steps can convert cart additions into completed purchases (directly tied to Objective 4).
- 3) Enhancing Product Transparency** – Introducing AR previews or 360-degree views, along with verified buyer reviews, can bridge the trust gap and reduce return rates.
- 4) Voice-Search and Vernacular Language Integration** – Making platforms available in Gujarati and Hindi can help non-English speakers participate more comfortably.

THREATS:

- 1) Over-Saturation of Common Platforms** - With many users buying from the same 2–3 platforms, new or local sellers may struggle for visibility.
- 2) Cybersecurity & Scam Risk** - Fake websites or misleading ads affect trust, especially among first-time users (a concern raised in SWOT-focused Objective 5).
- 3) Negative Reviews Impacting Brand Reputation** – A single viral negative review can heavily influence buyer behavior—even if the issue was isolated (Objective 3).
- 4) Delivery Delays During Peak Times** – Festival periods or city-specific events (like MSU exams/fairs) can cause delivery disruptions, leading to poor consumer experience.

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IBEF (2024).

India's Digital Consumers: A Demographic Shift.
McKinsey & Company (2023)

R-CODES :-

Goodman-Kruskal Test for Ordinal Categorical Variables : Test 1

```
library(MESS)

## Warning: package 'MESS' was built under R version 4.3.3|

gender=c("Female", "Male")
web_des_imp=c("Not Im.", "Some. Im.", "Important", "Very Im.")

m1=matrix(c(10,22,62,53,51,44,36,45), nrow=2, ncol=4, byrow=F)
rownames(m1)=gender
colnames(m1)=web_des_imp

gktest_1=gkgamma(m1,conf.level = 0.95)
gktest_1
```

Goodman-Kruskal's gamma for ordinal categorical data

```
data:  m1
Z = -0.14955, p-value = 0.8811
95 percent confidence interval:
 -0.1809722  0.1553120
sample estimates:
Goodman-Kruskal's gamma
      -0.01283011
```

Goodman-Kruskal Test for Ordinal Categorical Variables : Test 2

```
library(MESS)

## Warning: package 'MESS' was built under R version 4.3.3

gender=c("Female", "Male")
pro_des_imp=c("Not Us.", "Some. Us.", "Useful", "Very Us.")

m2=matrix(c(11,10,40,34,54,77,54,43), nrow=2, ncol=4, byrow=F)
rownames(m2)=gender
colnames(m2)=pro_des_imp

gktest_1=gkgamma(m2,conf.level = 0.95)
gktest_1|
```

Goodman-Kruskal's gamma for ordinal categorical data

```
data: m2
Z = -0.30958, p-value = 0.7569
95 percent confidence interval:
 -0.1997738  0.1452664
sample estimates:
Goodman-Kruskal's gamma
      -0.02725367
```

Goodman-Kruskal Test for Ordinal Categorical Variables : Test 3

```
library(MESS)

## Warning: package 'MESS' was built under R version 4.3.3

income=c("N/A", "VL", "L", "M", "H", "VH")
mon_spe=c("A", "B", "C", "D")

m3=matrix(c(20,9,8,4,21,15,5,3,30,39,21,11,18,40,15,13,8,8,6,10,5,4,6,4), nrow=6, ncol=4, byrow=T)
rownames(m3)=income
colnames(m3)=mon_spe

gktest_1=gkgamma(m3,conf.level = 0.95)
gktest_1
```

Goodman-Kruskal's gamma for ordinal categorical data

data: m3

$Z = 3.7954$, p-value = 0.0001474

95 percent confidence interval:

0.1175200 0.3623532

sample estimates:

Goodman-Kruskal's gamma

0.2399366

Goodman-Kruskal Test for Ordinal Categorical Variables : Test 4

```
library(MESS)

## Warning: package 'MESS' was built under R version 4.3.3

age=c("18-24","25-34","35-44","45-54","55-64","65 & Above")
rev_rea=c("Never","Sometimes","Always")

m4=matrix(c(8,22,19,7,39,80,5,23,71,4,7,13,3,6,13,1,1,1), nrow=6, ncol=3, byrow=T)
rownames(m4)=age
colnames(m4)=rev_rea

gktest_1=gkgamma(m4,conf.level = 0.95)
gktest_1|
```

Goodman-Kruskal's gamma for ordinal categorical data

```
data: m4
Z = 1.7564, p-value = 0.07901
95 percent confidence interval:
 -0.01756038  0.31235383
sample estimates:
Goodman-Kruskal's gamma
      0.1473967
```